

Konfigurasi DHCP Server pada Ubuntu Server

1. Pastikan VM mempunyai 2 NIC/Network Adapter. Adapter pertama menggunakan bridge, ini digunakan untuk mengkoneksikan ubuntu server ke internet, sehingga ip pada Network adapter ini akan secara otomatis diberikan (Dynamic). Adapter kedua menggunakan custom (contohnya VMnet1) dan ip pada adapter ini akan di setting secara manual (Static).

Device	Summary
Memory	2 GB
Processors	2
Hard Disk (SCSI)	40 GB
CD/DVD (SATA)	Using file C:\Users\helmi hut...
Network Adapter	Bridged (Automatic)
Network Adapter 2	Custom (VMnet1)
USB Controller	Present
Sound Card	Auto detect
Display	Auto detect

2. Cek ip nya, harus mendapatkan IP untuk NIC nya. Jika belum ada maka perlu di setting ip secara static. Pergi ke direktori /etc/netplan. Lalu edit file .yaml yang ada pada direktori itu.

```
GNU nano 7.2 50-cloud-init.yaml
network:
  version: 2
  ethernets:
    ens33:
      dhcp4: true
    ens37:
      dhcp4: no
      addresses:
        - 220.168.10.1/24
```

3. Kemudian terapkan konfigurasi tersebut dengan menegtikkan perintah berikut.

```
sudo netplan apply
```

4. Cek ip address nya kembali dan pastikan sudah ada.

```
ip a
```

```
nastarnas@nastarnas:/etc/netplan$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5d:06:3a brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.208.134/24 metric 100 brd 192.168.208.255 scope global dynamic ens33
        valid_lft 1752sec preferred_lft 1752sec
    inet6 fe80::20c:29ff:fe5d:63a/64 scope link
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5d:06:44 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    inet 220.168.10.1/24 brd 220.168.10.255 scope global ens37
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe5d:644/64 scope link
        valid_lft forever preferred_lft forever
nastarnas@nastarnas:/etc/netplan$ _
or press Ctrl+G.
```

5. Install DHCP Server menggunakan perintah

```
sudo apt install isc-dhcp-server
```

```
ce324003@ce3-server:~$ sudo apt install isc-dhcp-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  isc-dhcp-common
Suggested packages:
  isc-dhcp-server-ldap policycoreutils
The following NEW packages will be installed:
  isc-dhcp-common isc-dhcp-server
0 upgraded, 2 newly installed, 0 to remove and 144 not upgraded.
Need to get 1,281 kB of archives.
After this operation, 4,281 kB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

6. Konfigruasi `isc-dhcp-server`. Untuk pengkonfigurasian dilakukan pada file `/etc/dhcp/dhcpd.conf`. Cari pada baris `# A slightly different configuration for an internal subnet`. Kemudian hapus karakter `#` pada konfigurasi dibawahnya. Menjadi seperti ini.

```
sudo nano /etc/dhcp/dhcpd.conf
```

6. Tentukan interface mana yang akan menjalankan DHCP dengan mengkonfigurasi pada file `/etc/default/isc-dhcp-server`.

```
# A slightly different configuration for an internal subnet.
subnet 220.168.10.0 netmask 255.255.255.0 {
    range 220.168.10.10 220.168.10.100;
    option domain-name-servers ns1.internal.example.org;
    option domain-name "internal.example.org";
    option subnet-mask 255.255.255.224;
    option routers 220.168.10.1;
    option broadcast-address 220.168.10.225;
    default-lease-time 600;
    max-lease-time 7200;
}
```

```
sudo nano /etc/default/isc-dhcp-server
```

```
GNU nano 7.2 /etc/default/isc-dhcp-server *
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
# Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
# Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="ens34"
INTERFACESv6=""
```

7. Setelah konfigurasinya selesai restart service DHCP nya untuk menerapkan perubahan konfigurasi yang sudah dilakukan.

```
sudo systemctl restart isc-dhcp-server.service

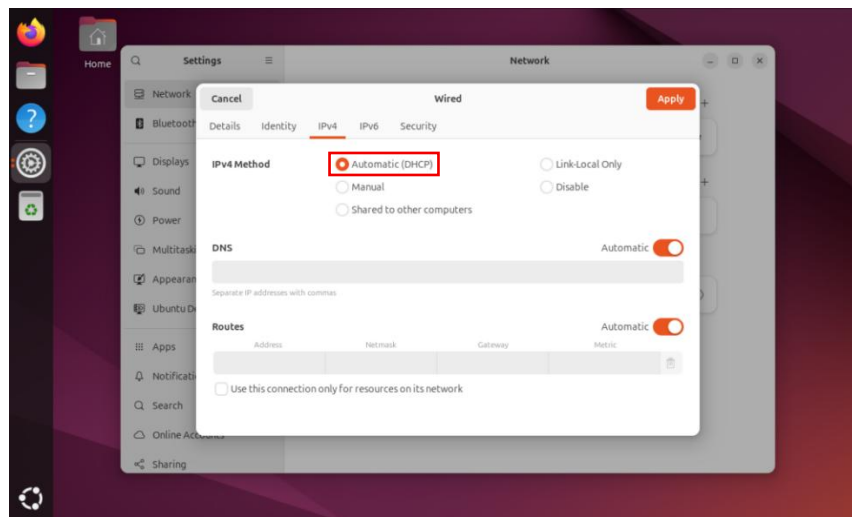
sudo systemctl status isc-dhcp-server
```

8. Sebelum melakukan pengujian kita harus melakukan pengaturan pada vm. Pengujian dilakukan dengan menggunakan vm lainnya (disini menggunakan ubuntu desktop). Untuk settingan network adaptornya menggunakan custom dan pastikan sama dengan adapter kedua pada ubuntu server. Jadi jika adapter kedua pada ubuntu server menggunakan custom: VMnet1 maka pada ubuntu

desktop juga wajib menggunakan custom: VMnet1. Ini artinya NIC 2 pada ubuntu server dengan NIC pada ubuntu desktop terhubung secara fisik.

Device	Summary
Memory	4 GB
Processors	2
Hard Disk (SCSI)	40 GB
CD/DVD 2 (SATA)	Using file C:\Users\helmi hut...
CD/DVD (SATA)	Using file autoinst.iso
Floppy	Using file autoinst.flp
Network Adapter	Custom (VMnet1)
USB Controller	Present
Sound Card	Auto detect
Display	Auto detect

9. Untuk melakukan pengujian, pastikan pengaturan ipv4 nya menggunakan Automatic bukan Manual, Shared to other computers, Link-Local Only, dan Disable.



10. Buka terminal pada ubuntu desktop dan ketikkan perintah ip a untuk melihat ip addressnya. Pastikan untuk ip networknya sama dengan nic network pada interface 2 pada ubuntu server.

```
nastarnas@nastarnas-VMware-Virtual-Platform: ~/Desktop
nastarnas@nastarnas-VMware-Virtual-Platform:~/Desktop$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:17:dd:c9 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    inet 220.168.10.10/24 brd 220.168.10.255 scope global dynamic noprefixroute ens37
        valid_lft 536sec preferred_lft 536sec
    inet6 fe80::87ef:d516:de78:7c7e/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
nastarnas@nastarnas-VMware-Virtual-Platform:~/Desktop$
```

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Device	Summary
Memory	2 GB
Processors	2
Hard Disk (SCSI)	40 GB
CD/DVD (SATA)	Using file C:\Users\helmi hut...
Network Adapter	Bridged (Automatic)
Network Adapter 2	Custom (VMnet1)
USB Controller	Present
Sound Card	Auto detect
Display	Auto detect

Gambar 1 Setingan VM Ubuntu Server

2. Cek ip nya, harus mendapatkan IP untuk NIC nya. Jika belum ada maka perlu di setting ip secara static. Pergi ke direktori /etc/netplan. Lalu edit file .yaml yang ada pada direktori itu.

```
GNU nano 7.2 50-cloud-init.yaml
network:
  version: 2
  ethernets:
    ens33:
      dhcp4: true
    ens37:
      dhcp4: no
      addresses:
        - 220.168.10.1/24
```

3. Kemudian terapkan konfigurasi tersebut dengan menegtikkan perintah berikut.

```
sudo netplan apply
```

4. Cek ip address nya kembali dan pastikan sudah ada.

```
ip a
```

```
nastarnas@nastarnas:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5d:06:3a brd ff:ff:ff:ff:ff:ff
    altnam enp2s1
    inet 192.168.208.134/24 metric 100 brd 192.168.208.255 scope global dynamic ens33
        valid_lft 1726sec preferred_lft 1726sec
    inet6 fe80::20c:29ff:fe5d:63a/64 scope link
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5d:06:44 brd ff:ff:ff:ff:ff:ff
    altnam enp2s5
    inet 220.168.10.1/24 brd 220.168.10.255 scope global ens37
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe5d:644/64 scope link
        valid_lft forever preferred_lft forever
nastarnas@nastarnas:~$
```

5. Install DHCP Server menggunakan perintah

```
sudo apt install isc-dhcp-server
```

6. Konfigurasi isc-dhcp-server. Untuk pengkonfigurasian dilakukan pada file `/etc/dhcp/dhcpd.conf`. Cari pada baris `# A slightly different configuration for an internal subnet`. Kemudian hapus karakter `#` pada konfigurasi dibawahnya. Menjadi seperti ini.

```
sudo nano /etc/dhcp/dhcpd.conf
```

```
# A slightly different configuration for an internal subnet.
subnet 220.168.10.0 netmask 255.255.255.0 {
    range 220.168.10.10 220.168.10.100;
    option domain-name-servers ns1.internal.example.org;
    option domain-name "internal.example.org";
    option routers 220.168.10.1;
    option broadcast-address 220.168.10.225;
    default-lease-time 600;
    max-lease-time 7200;
}
```

6. Tentukan interface mana yang akan menjalankan DHCP dengan mengkonfigurasi pada file `/etc/default/isc-dhcp-server`.

```
sudo nano /etc/default/isc-dhcp-server
```

```
Home x server x Ubuntu 64-bit x
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
# Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
# Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="ens37"
INTERFACESv6=""
```

7. Setelah konfigurasinya selesai restart service DHCP nya untuk menerapkan perubahan konfigurasi yang sudah dilakukan.

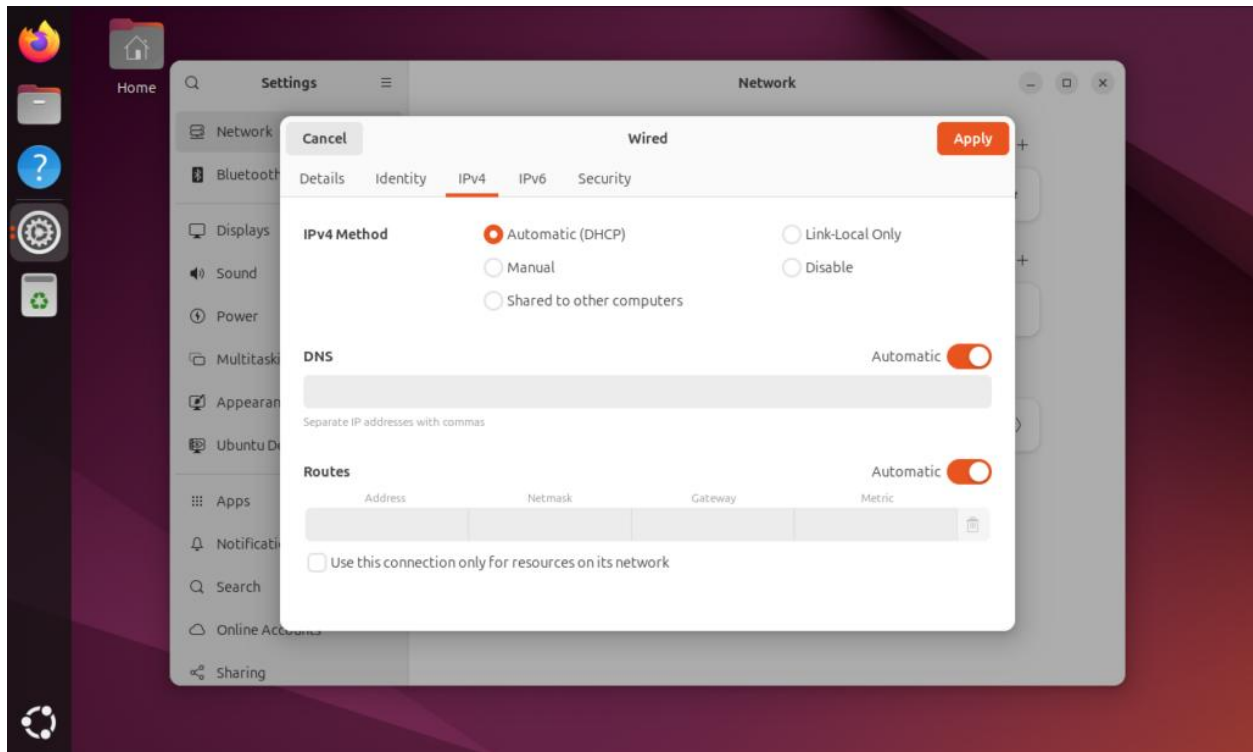
```
sudo systemctl status isc-dhcp-server
```

8. Sebelum melakukan pengujian kita harus melakukan pengaturan pada vm. Pengujian dilakukan dengan menggunakan vm lainnya (disini menggunakan ubuntu desktop). Untuk settingan network adapternya menggunakan custom dan pastikan sama dengan adapter kedua pada ubuntu server. Jadi jika adapter kedua pada ubuntu server menggunakan custom: VMnet1 maka pada ubuntu desktop juga wajib menggunakan custom: VMnet1. Ini artinya NIC 2 pada ubuntu server dengan NIC pada ubuntu desktop terhubung secara fisik.

Device	Summary
 Memory	4 GB
 Processors	2
 Hard Disk (SCSI)	40 GB
 CD/DVD 2 (SATA)	Using file C:\Users\helmi hut...
 CD/DVD (SATA)	Using file autoinst.iso
 Floppy	Using file autoinst.flp
 Network Adapter	Custom (VMnet1)
 USB Controller	Present
 Sound Card	Auto detect
 Display	Auto detect

Gambar 2 Settingan VM Ubuntu Desktop

9. Untuk melakukan pengujian, pastikan pengaturan ipv4 nya menggunakan Automatic bukan Manual, Shared to other computers, Link-Local Only, dan Disable.



10. Buka terminal pada ubuntu desktop dan ketikkan perintah `ip a` untuk melihat ip addressnya. Pastikan untuk ip networknya sama dengan ip network pada interface 2 pada ubuntu server.

Konfigurasi DNS Pada Ubuntu Server

1. Update dan upgrade ubuntu server anda sebelum menginstall dan memulai konfigurasi DNS server pada ubuntu server anda dengan command ***sudo apt update && sudo apt upgrade -y***

```
nastarnas@nastarnas:~$ sudo apt update && sudo apt upgrade -y
Hit:1 http://id.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://id.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://id.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following upgrades have been deferred due to phasing:
  ubuntu-pro-client ubuntu-pro-client-l10n
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
nastarnas@nastarnas:~$
```

2. Setelah selesai update dan upgrade ubuntu server lakukan instalasi bind9 yang merupakan software DNS server paling umum digunakan di Linux dengan command ***sudo apt install bind9***. Tampilan dibawah Adalah tampilan ketika sudah berhasil install bind9.

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
nastarnas@nastarnas:~$ sudo apt install bind9
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
bind9 is already the newest version (1:9.18.39-0ubuntu0.24.04.3).
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
nastarnas@nastarnas:~$
```

3. Cek status bind9 dengan command berikut :

sudo systemctl start bind9

sudo systemctl enable bind9

sudo systemctl status bind9

```
nastarnas@nastarnas:~$ sudo systemctl status named
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-04-23 07:24:34 UTC; 9min ago
     Docs: man:named(8)
  Main PID: 10813 (named)
    Status: "running"
     Tasks: 8 (limit: 2205)
  Memory: 23.8M (peak: 24.3M)
     CPU: 224ms
   CGroup: /system.slice/named.service
           └─10813 /usr/sbin/named -f -u bind

Apr 23 07:24:34 nastarnas named[10813]: network unreachable resolving './DNSKEY/IN': 2001:500:9f::42#53
Apr 23 07:24:34 nastarnas named[10813]: network unreachable resolving './NS/IN': 2001:500:9f::42#53
Apr 23 07:24:34 nastarnas named[10813]: network unreachable resolving './DNSKEY/IN': 2001:500:1::53#53
Apr 23 07:24:34 nastarnas named[10813]: network unreachable resolving './NS/IN': 2001:500:1::53#53
Apr 23 07:24:34 nastarnas named[10813]: managed-keys-zone: Initializing automatic trust anchor management for zone '.'; DNSKEY ID 20326 is now trusted, waiving
Apr 23 07:24:34 nastarnas named[10813]: managed-keys-zone: Initializing automatic trust anchor management for zone '.'; DNSKEY ID 38696 is now trusted, waiving
Apr 23 07:33:25 nastarnas named[10813]: no longer listening on 192.168.208.134#53
Apr 23 07:33:37 nastarnas named[10813]: listening on IPv4 interface ens33, 192.168.208.134#53
Apr 23 07:33:51 nastarnas named[10813]: no longer listening on 192.168.208.134#53
Apr 23 07:33:57 nastarnas named[10813]: listening on IPv4 interface ens33, 192.168.208.134#53
lines 1-22/22 (END)
nastarnas@nastarnas:~$
```

4. Masuk ke file konfigurasi DNS dengan command *sudo nano /etc/bind/named.conf.options* yang merupakan direktori konfigurasi DNS server. Lakukan konfigurasi pada file tersebut

IP yang digunakan Adalah IP DNS yang sudah kita subnetting di DHCP server

```
options {  
  
    directory "/var/cache/bind";  
  
    forwarders {  
        8.8.8.8;  
        8.8.4.4;  
    };  
  
    forward only;  
    dnssec-validation auto;  
  
    allow-query { any; };  
  
    recursion yes;  
  
    allow-recursion { localhost; 220.168.10.0/24; };  
  
    listen-on { 127.0.0.1; 220.168.10.11; };  
  
    listen-on-v6 { any; };  
};
```

Penjelasan :

- forwarders — jika client tanya google.com dan DNS lokal tidak tahu, pertanyaan diteruskan ke Google DNS
- allow-query — hanya perangkat di jaringan 192.168.10.0/24 yang bisa bertanya ke DNS ini pada kasus ini saya buat any biar bisa terima query dari mana saja.
- listen-on — DNS hanya aktif mendengarkan di IP server sendiri dan loopback, bukan interface lain

5. Akses file konfigurasi bind untuk mendaftarkan zona yaitu forward dan reverse dengan command ***sudo nano /etc/bind/named.conf.local***

Forward : fungsinya untuk menerjemahkan domain name → IP

Reverse : fungsinya untuk menerjemahkan IP → domain name

```
GNU nano 7.2 /etc/bind/named.conf.local
//
// Do any local configuration here
//
// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "zee.local" {
    type master;
    file "/etc/bind/zones/forward.zee";
    allow-update { none; };
};

zone "10.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/zones/reverse.zee";
    allow-update { none; };
};
```

Penjelasan :

- type master = server ini adalah sumber utama (bukan slave/secondary) untuk zona ini
- file = lokasi file zona yang akan dibuat di langkah berikutnya
- allow-update { none; } = tidak ada yang boleh mengubah zona ini secara dinamis (aman untuk jaringan lokal)
- Nama zona reverse 10.168.192.in-addr.arpa adalah standar DNS, ditulis dengan membalik 3 oktet pertama IP jaringan

6. Sekarang buatlah directory untuk membuat zona file tadi disini saya membuat nya di dalam /etc/bind/zones dengan command ***sudo mkdir /etc/bind/zones***

7. Selanjutnya buatlah file zone forward tadi dengan command ***sudo nano /etc/bind/zones/forward.josua*** dan ketikkan berikut :

```

$TTL      604800
@         IN      SOA      ns.zee.nm.del.ac.id. admin.zee.nm.del.ac.id. (
                                2025041202          ; Serial
                                604800              ; Refresh
                                86400               ; Retry
                                2419200             ; Expire
                                604800 )           ; Negative Cache TTL
;
@         IN      NS       ns.zee.nm.del.ac.id.
@         IN      A        220.168.10.1
ns        IN      A        220.168.10.1
www       IN      A        220.168.10.1
mail      IN      A        220.168.10.1

```

Penjelasan :

2025041201 ; Serial = format YYYYMMDDNN, naikan setiap edit

3600 ; Refresh = seberapa sering slave cek update (1 jam)

1800 ; Retry = jika refresh gagal, coba lagi tiap 30 menit

604800 ; Expire = data dianggap kadaluarsa setelah 1 minggu

86400) ; Negative TTL = cache "tidak ditemukan" selama 1 hari

<pre>@ IN NS ns1.josua.local. // Ini adalah Name Server (NS record).</pre> Fungsinya: Menentukan bahwa DNS utama untuk zone josua.local adalah ns1.josua.local.
--

<pre>ns1 IN A 192.168.10.10 // Ini adalah A record (Address Record).</pre> Fungsinya: Menghubungkan nama host ns1.josua.local ke IP fisik.

8. Selanjutnya buat file zone reverse tadi dengan command `dig` dan ketikan berikut :

```
Home X server X Ubuntu 64-bit X
; BIND reverse data file for 220.168.10.x
;
$TTL      604800
@         IN      SOA      ns.zee.nm.del.ac.id. admin.zee.nm.del.ac.id. (
                                2025041201      ; Serial
                                604800           ; Refresh
                                86400            ; Retry
                                2419200          ; Expire
                                604800 )         ; Negative Cache TTL
;
@         IN      NS       ns.zee.nm.del.ac.id.
1         IN      PTR      ns.zee.nm.del.ac.id.
```

Penjelasan :

Tidak jauh berbeda dengan forward ini adalah konfigurasi mengubah ip ke DNS

1	IN	PTR	ns1.josua.local	// 1 disini merupakan ip terakhir dari IP DNS kita
---	----	-----	-----------------	--

9. Kemudian lakukan pengecekan syntex konfigurasi dengan command berikut :

```
nastarnas@nastarnas:~$ sudo named-checkzone 10.168.220.in-addr.arpa /etc/bind/zones/reverse.zee
zone 10.168.220.in-addr.arpa/IN: loaded serial 2025041201
OK
nastarnas@nastarnas:~$
```

Pastikan semua zona ok jika terjadi eror maka cek file zona anda baik forward atau reverse.

10. Selanjutnya restart dan aktifkan bind9 agar semua konfigurasi kita tadi terbaca kemudian cek Kembali status bind9

```

josuamanurung@josua:~$ sudo systemctl restart bind9
josuamanurung@josua:~$ sudo systemctl enable bind9
Failed to enable unit: Refusing to operate on alias name or linked unit file: bind9.service
josuamanurung@josua:~$ sudo systemctl status bind9
* named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-04-14 03:59:47 UTC; 19s ago
     Docs: man:named(8)
    Main PID: 8119 (named)
      Status: "running"
    Tasks: 6 (limit: 4544)
   Memory: 22.3M (peak: 22.8M)
      CPU: 109ms
   CGroup: /system.slice/named.service
           └─8119 /usr/sbin/named -f -u bind

Apr 14 03:59:47 Josua named[8119]: zone 0.in-addr.arpa/IN: loaded serial 1
Apr 14 03:59:47 Josua named[8119]: zone Josua.local/IN: loaded serial 2025041201
Apr 14 03:59:47 Josua named[8119]: zone localhost/IN: loaded serial 2
Apr 14 03:59:47 Josua named[8119]: zone 127.in-addr.arpa/IN: loaded serial 1
Apr 14 03:59:47 Josua named[8119]: zone 255.in-addr.arpa/IN: loaded serial 1
Apr 14 03:59:47 Josua named[8119]: all zones loaded
Apr 14 03:59:47 Josua named[8119]: running
Apr 14 03:59:47 Josua systemd[1]: Started named.service - BIND Domain Name Server.
Apr 14 03:59:48 Josua named[8119]: managed-keys-zone: Key 20326 for zone . is now trusted (acceptance timer complete)
Apr 14 03:59:48 Josua named[8119]: managed-keys-zone: Key 38696 for zone . is now trusted (acceptance timer complete)
josuamanurung@josua:~$ _

```

11. Selanjutnya buka port DNS di firewall dengan command berikut

sudo firewall-cmd --add-port=53/tcp --permanent

sudo firewall-cmd --add-port=53/udp --permanent

sudo firewall-cmd --reload

Port 53 adalah port standar DNS, perlu dibuka untuk TCP dan UDP karena keduanya digunakan.

12. Pengujian dari server

a. Command ***nslookup domain_anda***

```

nastarnas@nastarnas:/$ nslookup zee.local
Server:          192.168.10.1
Address:         192.168.10.1#53

Name:   zee.local
Address: 192.168.10.1

```

b. Command ***dig domain_anda***

```

nastarnas@nastarnas:/$ dig www.zee.local

; <<>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<>> www.zee.local
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 21931
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: d2584a4adfb25e1e0100000069ece844de9c1a0ab13e63ec (good)
;; QUESTION SECTION:
;www.zee.local.                IN      A

;; ANSWER SECTION:
www.zee.local.                604800  IN      A      192.168.10.1

;; Query time: 11 msec
;; SERVER: 192.168.10.1#53(192.168.10.1) (UDP)
;; WHEN: Sat Apr 25 16:13:56 UTC 2026
;; MSG SIZE rcvd: 86

nastarnas@nastarnas:/$

```


Instalasi Web Server

1. Untuk melakukan tes dns melalui browser kita perlu melakukan instalasi web server disini saya menggunakan web server 1 dengan command berikut ***sudo apt install apache2*** tampilan dibawah merupakan tampilan ketika sudah berhasil menginstall apache2

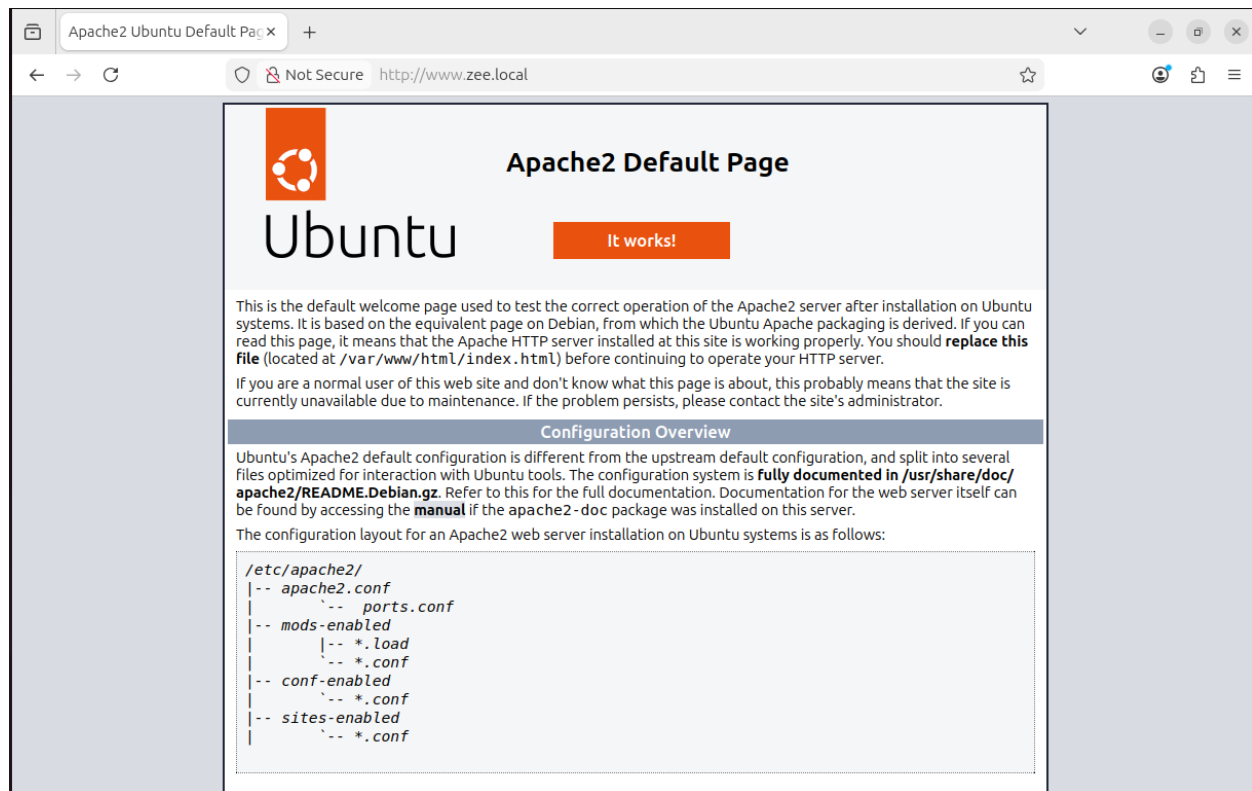
```
nastarnas@nastarnas:/$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
```

2. Selanjutnya cek status apache2 dengan command ***sudo systemctl status apache2***

Jika belum aktif gunakan command berikut ***sudo systemctl start apache2*** dan jika ingin menjalankan apache otomatis saat server aktif/restart gunakan command ***sudo systemctl enable apache2***

```
nastarnas@nastarnas:/$ sudo systemctl status apache2
• apache2.service - The Apache HTTP Server
  Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
  Active: active (running) since Sat 2025-04-25 16:28:55 UTC; 7s ago
  Docs: https://httpd.apache.org/docs/2.4/
  Process: 277079 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 277084 (apache2)
  Tasks: 55 (limit: 4543)
  Memory: 5.2M (peak: 5.4M)
  CPU: 268ms
  CGroup: /system.slice/apache2.service
          └─277084 /usr/sbin/apache2 -k start
            └─277085 /usr/sbin/apache2 -k start
              └─277087 /usr/sbin/apache2 -k start

Apr 25 16:28:55 nastarnas systemd[1]: Starting apache2.service - The Apache HTTP Server...
Apr 25 16:28:55 nastarnas apachectl[277081]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set t
Apr 25 16:28:55 nastarnas systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-17/17 (END)
nastarnas@nastarnas:/$
```



IP Client

```
2: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:17:dd:c9 brd ff:ff:ff:ff:ff:ff
    altnam enp2s5
    inet 192.168.10.10/27 brd 192.168.10.31 scope global dynamic noprefixroute ens37
        valid_lft 345sec preferred_lft 345sec
    inet6 fe80::87ef:d516:de78:7c7e/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

IP Ubuntu Server

```
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:4d:5f:64 brd ff:ff:ff:ff:ff:ff
    altnam enp2s2
    inet 192.168.10.1/24 brd 192.168.10.255 scope global ens34
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe4d:5f64/64 scope link
        valid_lft forever preferred_lft forever
```

Bisa kita lihat client bisa mengakses domain anda karena mereka berada pada satu IP yaitu di 192.168.10.x

Ringkasan Error :

1. Jika akses web anda menggunakan IP bisa tetapi menggunakan domain anda tidak bisa ada kesalahan konfigurasi di bagian zona forward atau anda bisa tambahkan file berikut di direktori

/etc/apache2/sites-available/domain.local.conf

```
<VirtualHost *:80>
  ServerName www.josua.local
  DocumentRoot /var/www/html

  <Directory /var/www/html>
    AllowOverride All
    Require all granted
  </Directory>
</VirtualHost>
```

Kemudian aktifkan dengan command ***sudo a2ensite bram.local.conf dan restart apache2***

2. jika saat nslookup atau dig tidak bisa untuk testing DNS ganti nameserver dan search pada file /etc/resolv.conf
3. Jika server not found saat akses domain dan client daftarkan domain anda dan IP domain anda pada konfigurasi DHCP di direktori /etc/dhcp/dhcp.d
4. jika eror saat restart bind atau apache2 ada kesalahan di file konfigurasi untuk cek eror ketik command ***journalctl -xe***

Instalasi & Konfigurasi NAT di Ubuntu Server

1. Untuk melakukan konfigurasi NAT di ubuntu server agar client mendapatkan akses internet kita perlu cek ip address kita dan mengetahui NIC mana yang ke internet dan NIC mana yang ke local pada vm saya menggunakan ens33 untuk internet dan ens37 untuk local jadi NAT berfungsi untuk menghubungkan kedua NIC ini agar ens37 bisa mengakses niternet melalui ens33.

```
nastarnas@nastarnas:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5d:06:3a brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.208.134/24 metric 100 brd 192.168.208.255 scope global dynamic ens33
        valid_lft 1726sec preferred_lft 1726sec
    inet6 fe80::20c:29ff:fe5d:63a/64 scope link
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5d:06:44 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    inet 220.168.10.1/24 brd 220.168.10.255 scope global ens37
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe5d:644/64 scope link
        valid_lft forever preferred_lft forever
nastarnas@nastarnas:~$
```

2. Selanjutnya set konfigurasi di interface LAN edit konfigurasi netplan dengan command ***sudo nano /etc/netplan/00-installer-config.yaml*** (file .yaml bisa saja berbeda tiap laptop) kemudian edit agar konfigurasi seperti pada gambar

```
GNU nano 7.2 /etc/netplan/50-cloud-init.yaml
network:
  version: 2
  ethernet:
    ens33:
      dhcp4: true

    ens37:
      - dhcp4: no
        addresses:
          - 192.168.10.1/24
```

3. Kemudian ketik command ***sudo netplan apply*** agar konfigurasi netplan kita tadi berjalan

```
josuamanurung@josua:~$ sudo netplan apply
josuamanurung@josua:~$
```

4. Selanjutnya kita aktifkan IP Forwarding nya di directory *sudo nano/etc/sysctl.conf* kemudian uncomment bagian ini sesuai pada gambar dibawah

```
# Uncomment the next line to enable TCP/IP SYN cookies
# See http://lwn.net/Articles/277146/
# Note: This may impact IPv6 TCP sessions too
#net.ipv4.tcp_syncookies=1

# Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=1

# Uncomment the next line to enable packet forwarding for IPv6
```

5. selanjutnya konfigurasi iptables (NAT/Masquerade) dengan command seperti gambar dibawah

```
nastarnas@nastarnas:/$ sudo iptables -t nat -A POSTROUTING -o ens33 -j MASQUERADE
nastarnas@nastarnas:/$ sudo iptables -A FORWARD -i ens34 -o ens33 -j ACCEPT
nastarnas@nastarnas:/$
```

```
nastarnas@nastarnas:/$ sudo iptables -A FORWARD -i ens33 -o ens34 -m state --state RELATED,ESTABLISHED -j ACCEPT
nastarnas@nastarnas:/$
```

6. Selanjutnya simpan agar perssisten setelah reboot

```
nastarnas@nastarnas:/$ sudo apt install iptables-persistent -y
Reading package lists... 0%
```

```
nastarnas@nastarnas:/$ sudo netfilter-persistent save
run-parts: executing /usr/share/netfilter-persistent/plugins.d/15-ip4tables save
# Warning: iptables-legacy tables present, use iptables-legacy-save to see them
run-parts: executing /usr/share/netfilter-persistent/plugins.d/25-ip6tables save
# Warning: ip6tables-legacy tables present, use ip6tables-legacy-save to see them
nastarnas@nastarnas:/$ sudo netfilter-persistent reload
run-parts: executing /usr/share/netfilter-persistent/plugins.d/15-ip4tables start
run-parts: executing /usr/share/netfilter-persistent/plugins.d/25-ip6tables start
nastarnas@nastarnas:/$
```

7. Kemudian restart dhcp server dengan command *sudo isc-dhcp-server* dan cek status dengan command *sudo status isc-dhcp-server* pastikan tidak ada eror dan pastikan dhcp server berjalan dengan lancar

8. selanjutnya akses dari client dan pastikan ip nya satu jaringan dengan server kemudian buka browser anda sudah bisa mengakses apapun seperti youtube.com atau website lainnya

Ringkasan Error :

1. Jika tidak bisa mengakses jaringan dari client test dengan ping dhcp server anda contoh : ping 192.168.10.1

2. Kemudian untuk memastikan bisa mengakses jaringan silahkan ping google.com atau [ing 8.8.8.8 dari client

3. Jika hanya bisa mengakses domain tidak bisa mengakses kalimat random silahkan tambahkan di konfigurasi dhcp server anda bagian option domain-name-server 8.8.8.8, 8.8.4.4;

Instalasi & Konfigurasi FTP Server di Ubuntu Server

1. Update vmware anda menggunakan command ***sudo apt update*** kemudian setelah selesai update unduh ubuntu server dengan commang ***sudo apt install vsftpd***.

```
nastarnas@nastarnas:/$ sudo apt update
Hit:1 http://id.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://id.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://id.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
nastarnas@nastarnas:/$ sudo apt install vsftpd -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libwrap0
The following NEW packages will be installed:
  libwrap0 vsftpd
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 168 kB of archives.
After this operation, 422 kB of additional disk space will be used.
0% [Waiting for headers]_
```

2. Aktifkan service fpt dengan command ***sudo systemctl start vsftpd*** dan jika ingin otomatis hidup saat restart / booting machine gunakan command ***sudo systemctl enable vsftpd***

```
nastarnas@nastarnas:/$ sudo systemctl start vsftpd
nastarnas@nastarnas:/$ sudo systemctl enable vsftpd
Synchronizing state of vsftpd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable vsftpd
nastarnas@nastarnas:/$ sudo systemctl status vsftpd
• vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-04-25 16:57:18 UTC; 19s ago
     Main PID: 320478 (vsftpd)
       Tasks: 1 (limit: 4543)
      Memory: 716.0K (peak: 856.0K)
         CPU: 5ms
        CGroup: /system.slice/vsftpd.service
                └─320478 /usr/sbin/vsftpd /etc/vsftpd.conf

Apr 25 16:57:18 nastarnas systemd[1]: Starting vsftpd.service - vsftpd FTP server...
Apr 25 16:57:18 nastarnas systemd[1]: Started vsftpd.service - vsftpd FTP server.
nastarnas@nastarnas:/$
```

3. Periksa status dari FPT server dengan commang ***sudo systemctl status vsftpd*** dan pastikan status nya running

```
nastarnas@nastarnas:/$ sudo systemctl start vsftpd
nastarnas@nastarnas:/$ sudo systemctl enable vsftpd
Synchronizing state of vsftpd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable vsftpd
nastarnas@nastarnas:/$ sudo systemctl status vsftpd
• vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-04-25 16:57:18 UTC; 19s ago
     Main PID: 320478 (vsftpd)
       Tasks: 1 (limit: 4543)
      Memory: 716.0K (peak: 856.0K)
         CPU: 5ms
        CGroup: /system.slice/vsftpd.service
                └─320478 /usr/sbin/vsftpd /etc/vsftpd.conf

Apr 25 16:57:18 nastarnas systemd[1]: Starting vsftpd.service - vsftpd FTP server...
Apr 25 16:57:18 nastarnas systemd[1]: Started vsftpd.service - vsftpd FTP server.
nastarnas@nastarnas:/$
```

4. jika FTP server sudah berhasil berjalan selanjutnya masuk ke file konfigurasi FTP server directory berada di `/etc/vsftpd.conf` masuk dengan commang ***sudo nano /etc/vsftpd.conf***

```
# access.
secure_chroot_dir=/var/run/vsftpd/empty
#
# This string is the name of the PAM service vsftpd will use.
pam_service_name=vsftpd
#
# This option specifies the location of the RSA certificate to use for SSL
# encrypted connections.
rsa_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
rsa_private_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
ssl_enable=NO

#
# Uncomment this to indicate that vsftpd use a utf8 filesystem.
#utf8_filesystem=YES

allow_writeable_chroot=YES
```

Cari baris seperti ini dan sesuaikan dengan konfigurasi pada kolom

```
anonymous_enable=NO
local_enable=YES
write_enable=YES
chroot_local_user=YES
```

Bagian ini tambahkan di paling bawah pada konfigurasi !!

```
# Passive mode (penting agar client bisa konek dari luar)
pasv_enable=YES
pasv_min_port=10000
pasv_max_port=10100
```


5. Setelah selesai melakukan konfigurasi anda dapat restart FTP server dengan command ***sudo systemctl restart vsftpd*** dan cek status nya pastikan running jika gagal/eror bberarti ada kesalahan pada file konfigurasi.

```
nastarnas@nastarnas:/$ sudo systemctl restart vsftpd
nastarnas@nastarnas:/$ sudo systemctl status vsftpd
• vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-04-25 17:02:16 UTC; 13s ago
     Process: 328354 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SUCCESS)
    Main PID: 328356 (vsftpd)
      Tasks: 1 (limit: 4543)
     Memory: 720.0K (peak: 1.5M)
        CPU: 12ms
      CGroup: /system.slice/vsftpd.service
              └─328356 /usr/sbin/vsftpd /etc/vsftpd.conf

Apr 25 17:02:16 nastarnas systemd[1]: Starting vsftpd.service - vsftpd FTP server...
Apr 25 17:02:16 nastarnas systemd[1]: Started vsftpd.service - vsftpd FTP server.
nastarnas@nastarnas:/$
```

6. Selanjutnya kita akan membuat autentikasi untuk mengakses file di ftp server dengan command ***sudo adduser (namauser)*** selanjutnya akan masukan password dimana itu berguna untuk autentikasi saat ingin mengkases FTP server.

```
nastarnas@nastarnas:/$ sudo adduser ftpuser
info: Adding user `ftpuser' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `ftpuser' (1002) ...
info: Adding new user `ftpuser' (1002) with group `ftpuser (1002)' ...
info: Creating home directory `/home/ftpuser' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
Sorry, passwords do not match.
passwd: Authentication token manipulation error
passwd: password unchanged
Try again? [y/N] y
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for ftpuser
Enter the new value, or press ENTER for the default
   Full Name []: nastarnas
   Room Number []: 1
   Work Phone []: 1
   Home Phone []: 1
   Other []: 1
Is the information correct? [Y/n] y
info: Adding new user `ftpuser' to supplemental / extra groups `users' ...
info: Adding user `ftpuser' to group `users' ...
nastarnas@nastarnas:/$ sudo mkdir -p /home/ftpuser/ftpfile
```

7. Selanjutnya kita akan membuat folder yang akan diakses dan juga juga folder yang akan di upload dengan command ***sudo mkdir /home/ftpuser/ftpfile*** untuk file akses dan ***sudo mkdir /home/ftpuser/ftp/upload*** untuk file upload.

```
// Folder untuk Akses
mkdir /home/ftpuser/ftp
chmod 755 /home/ftpuser/ftp

//Folder untuk upload
mkdir /home/ftpuser/ftp/upload
chown ftpuser:ftpuser /home/ftpuser/ftp/upload
```

8. Atur permission wajib jika menggunakan root

```
sudo chown root:root /home/ftpuser
sudo chmod 755 /home/ftpuser
```

9. Buka firewall dengan port khusus untuk ftp yaitu port 21 dengan command ***sudo ufw allow 21*** dan ***sudo ufw allow 10000:10100/tcp***

```
root@josua:/home/josuamanurung# sudo ufw allow 21
Rules updated
Rules updated (v6)
root@josua:/home/josuamanurung# sudo ufw allow 10000:10100/tcp
Rules updated
Rules updated (v6)
root@josua:/home/josuamanurung# _
```

10. Kemudian akses dari client dengan IP yang sama adri DHCP buka terminal dan ketik **ftp (ip server)**

```
nastarnas@nastarnas-VMware-Virtual-Platform:~/Desktop$ ftp 192.168.10.1
Connected to 192.168.10.1.
220 (vsFTPd 3.0.5)
Name (192.168.10.1:nastarnas): ftpuser
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

atau jika ingin menggunakan yang memiliki GUI anda bisa menginstall filezilla di ubuntu client dengan command ***sudo apt install filezilla*** dan ketika sudah selesai akses filezilla dan masuka IP, username, password, berserta port untuk konek ke FTP server.

Instalasi & Konfigurasi Mail Server pada Ubuntu Server

1. Sebelum install mail server kita perlu memperhatikan bahwa dns Josua.local itu punya mail server . MX Record adalah penanda kemana email itu dikirim sehingga kita perlu mengedit file forward dns kita dengan menambahkan MX

```
GNU nano 7.2 /etc/bind/zones/forward.zee
$TTL 604800

@ IN SOA ns1.zee.local. admin.zee.local. (
    2025042501
    3600
    1800
    604800
    86400 )

@ IN NS ns1.zee.local.
@ IN A 192.168.10.1
ns1 IN A 192.168.10.1
www IN A 192.168.10.1
mail IN A 192.168.10.1
```

2. cek konfigurasi forward nya dengan command ***sudo named-checkzone josua.local /etc/bind/zones/forward.josua*** dan restart bind9 nya dengan command ***sudo systemctl restart bind9***

```
nastarnas@nastarnas:/$ sudo named-checkzone zee.local /etc/bind/zones/forward.zee
zone zee.local/IN: loaded serial 2025042502
OK
nastarnas@nastarnas:/$ _
```

3. verifikasi mail server dengan command dig MX Josua.local @192.168.10.1

```
nastarnas@nastarnas:/$ dig MX zee.local @192.168.10.1

; <>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <>> MX zee.local @192.168.10.1
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 36993
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 2

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: d7e0870fa9dd76dd0100000069ecf830a609fdb0280b1cc5 (good)
;; QUESTION SECTION:
;zee.local.                IN      MX

;; ANSWER SECTION:
zee.local.                604800  IN      MX      10 mail.zee.local.

;; ADDITIONAL SECTION:
mail.zee.local.          604800  IN      A       192.168.10.1

;; Query time: 1 msec
;; SERVER: 192.168.10.1#53(192.168.10.1) (UDP)
;; WHEN: Sat Apr 25 17:21:52 UTC 2026
;; MSG SIZE rcvd: 103

nastarnas@nastarnas:/$
```

4. Selanjutnya instalasi postfix dengan command `sudo apt install postfix -y` Postfix bertugas menerima dan mengirim email antar server maupun ke mailbox lokal. Gambar dibawah merupakan tampilan sudah berhasil install postfix.

```
root@josua:/home/josuamanurung# sudo apt install postfix -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
postfix is already the newest version (3.8.6-1build2).
0 upgraded, 0 newly installed, 0 to remove and 32 not upgraded.
root@josua:/home/josuamanurung# _
```

5. Masuk ke file konfigurasi postfix dengan command ***sudo nano /etc/postfix/main.cf*** dan mulai lah lakukan konfigurasi , Digambar dibawah saya menghapus semua komentar agar file konfigurasi lebih bersih dan mudah untuk diedit sesuaikan DNS dan network dengan milik kalian sendiri.

```

smtpd_banner = $myhostname ESMTP
biff = no
append_dot_mydomain = no
myhostname = mail.josua.local
mydomain = josua.local
myorigin = $mydomain
inet_interfaces = all
inet_protocols = ipv4
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
mynetworks = 127.0.0.0/8, 192.168.10.0/24
home_mailbox = Maildir/
message_size_limit = 10240000
smtpd_sasl_type = dovecot
smtpd_sasl_path = private/auth
smtpd_sasl_auth_enable = yes
smtpd_recipient_restrictions =
    permit_mynetworks,
    permit_sasl_authenticated,
    reject_unauth_destination

```

6. Kemudian restart postfix agar konfigurasi kita berjalan

```

root@josua:/home/josuamanurung# sudo systemctl restart postfix
root@josua:/home/josuamanurung# _

```

7. kemudian install dovecot ***sudo apt install dovecot-core dovecot-imapd -y*** yang berfungsi untuk menyimpan email dan melayani client yang ingin membaca email via protokol IMAP.

```

root@josua:/home/josuamanurung# sudo apt install dovecot-core dovecot-imapd -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
dovecot-core is already the newest version (1:2.3.21+dfsg1-2ubuntu6.3).
dovecot-imapd is already the newest version (1:2.3.21+dfsg1-2ubuntu6.3).
0 upgraded, 0 newly installed, 0 to remove and 32 not upgraded.
root@josua:/home/josuamanurung# _

```

8. kemudian buka file konfigurasi dovecot dengan command ***sudo nano etc/dovecot/dovecot.conf***

```
# Enable installed protocols
!include_try /usr/share/dovecot/protocols.d/*.protocol

# A comma separated list of IPs or hosts where to listen in for connections.
# "*" listens in all IPv4 interfaces, ":::" listens in all IPv6 interfaces.
# If you want to specify non-default ports or anything more complex,
# edit conf.d/master.conf.
listen = *, ::
```

9. konfigurasi Lokasi dari mail box dengan dengan command **sudo nano etc/dovecot/conf.d/10-mail.conf**

```
# <doc/wiki/MailLocation.txt>
#
mail_location = maildir:~/Maildir
```

10. Kemudian konfigurasi file **sudo nano /etc/dovecot/conf.d/10-auth.conf** konfigurasi ini berkaitan dengan autentikasi user

```
# Disable LOGIN command and all other plaintext authentications unless
# SSL/TLS is used (LOGINDISABLED capability). Note that if the remote IP
# matches the local IP (ie. you're connecting from the same computer), the
# connection is considered secure and plaintext authentication is allowed.
# See also ssl=required setting.
disable_plaintext_auth = no
```

```
# Space separated list of wanted authentication mechanisms:
#   plain login digest-md5 cram-md5 ntlm rpa apop anonymous gssapi otp
#   gss-spnego
# NOTE: See also disable_plaintext_auth setting.
auth_mechanisms = plain login
```

11. kemudian konfigurasi file **sudo nano /etc/dovecot/conf.d/10-master.conf** Dovecot membuat socket file di lokasi yang bisa dibaca Postfix, sehingga saat user login SMTP, Postfix bisa bertanya ke Dovecot "apakah password ini benar?".

```

service auth {
    # auth_socket_path points to this userdb socket by default. It's typically
    # used by dovecot-lda, doveadm, possibly imap process, etc. Users that have
    # full permissions to this socket are able to get a list of all usernames and
    # get the results of everyone's userdb lookups.
    #
    # The default 0666 mode allows anyone to connect to the socket, but the
    # userdb lookups will succeed only if the userdb returns an "uid" field that
    # matches the caller process's UID. Also if caller's uid or gid matches the
    # socket's uid or gid the lookup succeeds. Anything else causes a failure.
    #
    # To give the caller full permissions to lookup all users, set the mode to
    # something else than 0666 and Dovecot lets the kernel enforce the
    # permissions (e.g. 0777 allows everyone full permissions).
    unix_listener auth-userdb {
        mode = 0600
        user = dovecot
    }

    # Postfix smtp-auth
    unix_listener /var/spool/postfix/private/auth {
        mode = 0660
        user = postfix
        group = postfix
    }

    # Auth process is run as this user.
    #user = $default_internal_user
}

```

12. restart dovecot agar semua file konfigurasi kita berjalan

```

root@josua:/home/josuamanurung# sudo systemctl restart dovecot
root@josua:/home/josuamanurung#

```

13. kemudian tambahkan 2 user untuk melakukan pengiriman mail dengan command ***sudo adduser (namauser)*** dan otomatis menambahkan password

14. kemudian anda bisa mulai mengirim pesan dari user yang anda tambahkan disini saya membuat user Josua dan admin

```

root@josua:/home/josuamanurung# sudo telnet mail.josua.local 25
Trying 192.168.10.1...
Connected to mail.josua.local.
Escape character is '^]'.
mail from: josua@jos220 mail.josua.local ESMTP
quit
250 2.1.0 Ok
quit
221 2.0.0 Bye
Connection closed by foreign host.
root@josua:/home/josuamanurung# sudo telnet mail.josua.local 25
Trying 192.168.10.1...
Connected to mail.josua.local.
Escape character is '^]'.
220 mail.josua.local ESMTP
mail from: josua@josua@mail.josua.local
250 2.1.0 Ok
rcpt to: admin@josua.local
250 2.1.5 Ok
data
354 End data with <CR><LF>.<CR><LF>
Selamat Kuliah
.
250 2.0.0 Ok: queued as 0413E10AA
quit
221 2.0.0 Bye
Connection closed by foreign host.

```

15. kemudian kita bisa lihat mail Josua disisi admin

```
root@josua:/home/josuamanurung# sudo telnet mail.josua.local 110
Trying 192.168.10.1...
Connected to mail.josua.local.
Escape character is '^]'.
+OK Dovecot (Ubuntu) ready.
user admin
+OK
pass josuamanurung123
+OK Logged in.
stat
+OK 4 1974
retr 4
+OK 441 octets
Return-Path: <"josua@josua"@mail.josua.local>
X-Original-To: admin@josua.local
Delivered-To: admin@josua.local
Received: from unknown (unknown [192.168.10.1])
    by mail.josua.local (Postfix) with SMTP id 0413E10AA
    for <admin@josua.local>; Sun, 19 Apr 2026 06:07:17 +0000 (UTC)
Message-Id: <20260419060731.0413E10AA@mail.josua.local>
Date: Sun, 19 Apr 2026 06:07:17 +0000 (UTC)
From: "josua@josua"@mail.josua.local

Selamat Kuliah
.
```


Konfigurasi Web Server pada Ubuntu Server

Kali ini kita akan menginstall web server pada ubuntu server. Web yang akan kita install menggunakan bahasa php native dan menggunakan sql sebagai databsnya. Maka dari itu kita perlu menginstall apache2 dan mysql untuk lingkungan. Dan website yang akan di install akan di ambil atau di clone dari github.

1. Silahkan melakukan update ubuntu dengan mengetikkan perintah berikut.

```
sudo apt update
```

- ## 2. Install apache2.

```
sudo apt install apache2
```

```
nastarmas@nastarmas:/$ sudo apt install php libapache2-mod-php php-mysql mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libapache2-mod-php8.3 libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi10t64
  libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2
  libprotobuf-lite32t64 libltime-date-perl liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common
  mysql-server-8.0 mysql-server-core-8.0 php-common php8.3 php8.3-cli php8.3-common php8.3-mysql php8.3-opcache php8.3-readline
Suggested packages:
  php-pear libdata-dump-perl libipc-sharedcache-perl libio-compress-brotli-perl libbusiness-isbn-perl libregexp-ipv6-perl libwww-perl mailx tinyca
The following NEW packages will be installed:
  libapache2-mod-php libapache2-mod-php8.3 libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin
  libfcgi-perl libfcgi10t64 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl
  liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64 libtime-date-perl liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0
  mysql-client-core-8.0 mysql-common mysql-server mysql-server-8.0 mysql-server-core-8.0 php php-common php-mysql php8.3 php8.3-cli php8.3-common php8.3-mysql
  php8.3-opcache php8.3-readline
0 upgraded, 39 newly installed, 0 to remove and 0 not upgraded.
Need to get 34.7 MB of archives.
After this operation, 266 MB of additional disk space will be used.
Get:1 http://id.archive.ubuntu.com/ubuntu noble/amd64 mysql-common all 5.8+1.1.0build1 [6746 B]
Get:2 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-client-core-8.0 amd64 8.0.45-0ubuntu0.24.04.1 [2740 kB]
Get:3 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-client-8.0 amd64 8.0.45-0ubuntu0.24.04.1 [22.5 kB]
Get:4 http://id.archive.ubuntu.com/ubuntu noble/amd64 libevent-pthreads-2.1-7t64 amd64 2.1.12-stable-9ubuntu2 [7982 B]
Get:5 http://id.archive.ubuntu.com/ubuntu noble/main amd64 libmecab2 amd64 0.996-14ubuntu4 [201 kB]
Get:6 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 libprotobuf-lite32t64 amd64 3.21.12-8.2ubuntu0.3 [238 kB]
9% [6 libprotobuf-lite32t64 17.0 kB/238 kB 7%]
```

- ### 3. Install module php.

```
sudo apt install php libapache2-mod-php php-mysql
```

```
nastarnas@nastarnas:/$ sudo apt install php libapache2-mod-php php-mysql mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libapache2-mod-php8.3 libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi0t64
  libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2
  libprotobuf-lite32t64 libtimedate-perl liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common
  mysql-server-8.0 mysql-server-core-8.0 php-common php8.3 php8.3-cli php8.3-common php8.3-mysql php8.3-opcache php8.3-readline
Suggested packages:
  php-pear libdata-dump-perl libipc-sharedcache-perl libio-compress-brotli-perl libbusiness-isbn-perl libregexp-ipv6-perl libwww-perl mailx tinyca
The following NEW packages will be installed:
  libapache2-mod-php libapache2-mod-php8.3 libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin
  libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl
  liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64 libtimedate-perl liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0
  mysql-client-core-8.0 mysql-common mysql-server mysql-server-8.0 mysql-server-core-8.0 php php-common php-mysql php8.3 php8.3-cli php8.3-common php8.3-m
  php8.3-opcache php8.3-readline
0 upgraded, 39 newly installed, 0 to remove and 0 not upgraded.
Need to get 34.7 MB of archives.
After this operation, 266 MB of additional disk space will be used.
Get:1 http://id.archive.ubuntu.com/ubuntu noble/main amd64 mysql-common all 5.8+1.1.0build1 [6746 B]
Get:2 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-client-core-8.0 amd64 8.0.45-0ubuntu0.24.04.1 [2740 kB]
Get:3 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-client-8.0 amd64 8.0.45-0ubuntu0.24.04.1 [22.5 kB]
Get:4 http://id.archive.ubuntu.com/ubuntu noble/main amd64 libevent-pthreads-2.1-7t64 amd64 2.1.12-stable-9ubuntu2 [7982 B]
Get:5 http://id.archive.ubuntu.com/ubuntu noble/main amd64 libmecab2 amd64 0.996-14ubuntu4 [201 kB]
Get:6 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 libprotobuf-lite32t64 amd64 3.21.12-8.2ubuntu0.3 [238 kB]
9% [6 libprotobuf-lite32t64 17.0 kB/238 kB 7%]
```

3. Install mysql.

```
sudo apt install mysql-server
```

```
nastarnas@nastarnas:/$ sudo apt install php libapache2-mod-php php-mysql mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libapache2-mod-php8.3 libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi0t64
  libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2
  libprotobuf-lite32t64 libtimedate-perl liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common
  mysql-server-8.0 mysql-server-core-8.0 php-common php8.3 php8.3-cli php8.3-common php8.3-mysql php8.3-opcache php8.3-readline
Suggested packages:
  php-pear libdata-dump-perl libipc-sharedcache-perl libio-compress-brotli-perl libbusiness-isbn-perl libregexp-ipv6-perl libwww-perl mailx tinyca
The following NEW packages will be installed:
  libapache2-mod-php libapache2-mod-php8.3 libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin
  libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl
  liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64 libtimedate-perl liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0
  mysql-client-core-8.0 mysql-common mysql-server mysql-server-8.0 mysql-server-core-8.0 php php-common php-mysql php8.3 php8.3-cli php8.3-common php8.3-m
  php8.3-opcache php8.3-readline
0 upgraded, 39 newly installed, 0 to remove and 0 not upgraded.
Need to get 34.7 MB of archives.
After this operation, 266 MB of additional disk space will be used.
Get:1 http://id.archive.ubuntu.com/ubuntu noble/main amd64 mysql-common all 5.8+1.1.0build1 [6746 B]
Get:2 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-client-core-8.0 amd64 8.0.45-0ubuntu0.24.04.1 [2740 kB]
Get:3 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-client-8.0 amd64 8.0.45-0ubuntu0.24.04.1 [22.5 kB]
Get:4 http://id.archive.ubuntu.com/ubuntu noble/main amd64 libevent-pthreads-2.1-7t64 amd64 2.1.12-stable-9ubuntu2 [7982 B]
Get:5 http://id.archive.ubuntu.com/ubuntu noble/main amd64 libmecab2 amd64 0.996-14ubuntu4 [201 kB]
Get:6 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 libprotobuf-lite32t64 amd64 3.21.12-8.2ubuntu0.3 [238 kB]
9% [6 libprotobuf-lite32t64 17.0 kB/238 kB 7%]
```

4. Install git.

```
sudo apt install git -y
```

```
ce324003@ce3-server:~$ sudo apt install git -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 147 not upgraded.
ce324003@ce3-server:~$
```

5. Masuk ke direktori /var/www/html kemudian clone salah satu repostiroy dari github.

```
sudo git clone https://github.com/RafyMrX/TOKO-ONLINE-ROTI.git
```

```
ce324003@ce3-server:/var/www/html$ sudo git clone https://github.com/RafyMrX/TOKO-ONLINE-ROTI.git
Cloning into 'TOKO-ONLINE-ROTI'...
remote: Enumerating objects: 118, done.
remote: Counting objects: 100% (118/118), done.
remote: Compressing objects: 100% (101/101), done.
remote: Total 118 (delta 23), reused 101 (delta 15), pack-reused 0 (from 0)
Receiving objects: 100% (118/118), 4.31 MiB | 928.00 KiB/s, done.
Resolving deltas: 100% (23/23), done.
ce324003@ce3-server:/var/www/html$ _
```

6. Kemudian pindah semua isi dari direktori TOKO-ONLINE-ROTI tersebut ke direktori /var/www/html dengan menggunakan perintah berikut.

```
sudo mv TOKO-ONLINE-ROTI/* /var/www/html
```

7. Lalu kita masuk ke mysql untuk membuat databasenya. Lalu lihat database yang sudah dibuat.

```
ce324003@ce3-server:/var/www/html$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE dbpw192_18410100054;
Query OK, 1 row affected (0.01 sec)

mysql> SHOW DATABASES
->
+-----+
| Database |
+-----+
| dbpw192_18410100054 |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.01 sec)
```

8. Kemudian buat dan berikan previllage user untuk mengkases database tersebut.

```
mysql> CREATE USER 'helmi'@'localhost' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.02 sec)

mysql> GRANT ALL PRIVILEGES ON dbpw192_18410100054.* TO 'helmi'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.00 sec)
```

9. Keluar dari mysql kemudian import database dari direktori github yang sudah diclone tadi menggunakan user yang sudah dibuat.

```
mysql -u helmi -p dbpw192_18410100054 < /var/www/html/DATABASE/dbpw192_18410100054.sql
```

```
ce324003@ce3-server:/var/www/html$ mysql -u helmi -p dbpw192_18410100054 < /var/www/html/DATABASE/dbpw192_18410100054.sql
Enter password:
ce324003@ce3-server:/var/www/html$
```

10. Karena php perlu file untuk mengkoneksikan ke database maka ubah username dan password pada file koneksi tersebut. Pastikan username dan passwordnya sesuai dengan Langkah 8.

```
GNU nano 7.2 koneksi.php *
<?php
$conn = mysqli_connect("localhost", "helmi", "password123", "dbpw192_18410100054");
?>
```

11. Lalu cari ip ubuntu server (Network Adapter 1) pada browser host (misalnya google chrome). Makawebbsite yang sudah dibuat pada ubuntu server akan tampil.

PHP 8.3.6 - phpinfo()

Not Secure http://192.168.10.1/info.php

PHP Version 8.3.6



System	Linux nastarnas 6.8.0-110-generic #110-Ubuntu SMP PREEMPT_DYNAMIC Thu Mar 19 15:09:20 UTC 2026 x86_64
Build Date	Mar 20 2026 02:32:55
Build System	Linux
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php/8.3/apache2
Loaded Configuration File	/etc/php/8.3/apache2/php.ini
Scan this dir for additional .ini files	/etc/php/8.3/apache2/conf.d
Additional .ini files parsed	/etc/php/8.3/apache2/conf.d/10-mysqld.ini, /etc/php/8.3/apache2/conf.d/10-opcache.ini, /etc/php/8.3/apache2/conf.d/10-pdo.ini, /etc/php/8.3/apache2/conf.d/20-calendar.ini, /etc/php/8.3/apache2/conf.d/20-ctype.ini, /etc/php/8.3/apache2/conf.d/20-exif.ini, /etc/php/8.3/apache2/conf.d/20-ffi.ini, /etc/php/8.3/apache2/conf.d/20-fileinfo.ini, /etc/php/8.3/apache2/conf.d/20-ftp.ini, /etc/php/8.3/apache2/conf.d/20-gettext.ini, /etc/php/8.3/apache2/conf.d/20-iconv.ini, /etc/php/8.3/apache2/conf.d/20-mysqli.ini, /etc/php/8.3/apache2/conf.d/20-pdo_mysql.ini, /etc/php/8.3/apache2/conf.d/20-phar.ini, /etc/php/8.3/apache2/conf.d/20-posix.ini, /etc/php/8.3/apache2/conf.d/20-readline.ini, /etc/php/8.3/apache2/conf.d/20-shmop.ini, /etc/php/8.3/apache2/conf.d/20-sockets.ini, /etc/php/8.3/apache2/conf.d/20-sysvmsg.ini, /etc/php/8.3/apache2/conf.d/20-sysvsem.ini, /etc/php/8.3/apache2/conf.d/20-sysvshm.ini, /etc/php/8.3/apache2/conf.d/20-tokenizer.ini
PHP API	20230831
PHP Extension	20230831
Zend Extension	420230831
Zend Extension Build	API420230831,NTS
PHP Extension Build	API20230831,NTS
Debug Build	no
Thread Safety	disabled